Muhammad Ayaz

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OBJECTIVE

Enthusiastic researcher with a strong background in Computer Vision, Multimodal AI, and Deep Learning, with research experience spanning fire and smoke understanding, vision-language models, affective computing, medical AI, and disaster scene analysis. Interested in developing robust and deployable intelligent systems for real-world applications, with strong proficiency in Python, PyTorch, TensorFlow, and Keras.

PROFESSIONAL EXPERIENCE

Research Assistant *Mar 2025 - Present*
Graphics Realization Lab (GRLab), Chung-Ang University *Seoul, South Korea*
- Conduct research in computer vision, image segmentation, multimodal learning, and deep learning under Prof. Sanghyun Seo.
- Develop robust models for fire and smoke understanding, medical vision-language reasoning, and real-world visual perception.

Research Assistant *Sep 2022 - Dec 2024*
Digital Image Processing Laboratory, Islamia College University *Peshawar, Pakistan*
- Contributed to image classification, 2D and 3D object detection, image segmentation, autonomous driving, and video understanding projects.
- Supported dataset preparation, model training, experimental evaluation, and research documentation.

Course Assistant *May 2023 - Dec 2024*
Digital Image Processing Laboratory, Islamia College University *Peshawar, Pakistan*
- Assisted with courses in artificial intelligence, computer vision, machine learning, and deep learning.

Junior Researcher *Dec 2020 - 2022*
National Center for Big Data and Cloud Computing (NCBC) *Peshawar, Pakistan*
- Participated in data collection, annotation, preprocessing, and computer vision research tasks.

SELECTED PROJECTS

ROAD-R Challenge for NeurIPS 2023 *2023*
- Worked on autonomous-driving event understanding using the ROAD-R dataset and logical requirements.

MedVLM: Vision-Language Model for Holistic Medical Applications *2024 - 2025*
- Trained and evaluated multimodal models for medical visual question answering using X-rays, CT scans, and MRIs.
- Collaborated with researchers at MBZUAI and United Arab Emirates University.

Cone Detection for Formula Student Cars
- Developed and fine-tuned object detection models for cone-based vehicle navigation using the FSOCO dataset.

Fire and Smoke Instance Segmentation in Challenging Environments
- Developed vision models for accurate fire and smoke instance segmentation across indoor and outdoor scenes.
- Improved robustness under difficult lighting, occlusion, scale variation, and complex environmental conditions.

Anomaly Detection in Video Sequences
- Combined convolutional feature extraction with recurrent temporal modeling to identify abnormal video events.

EDUCATION

MS, Applied Arts and Engineering *2025 - Present*
Chung-Ang University (CAU), Graduate School of Applied Arts and Engineering *Seoul, South Korea*
- GPA: 4.33/4.5 (98%); current laboratory: Graphics Realization Lab (GRLab).

BS Computer Science *2019 - 2023*
The University of Agriculture Peshawar (AUP) *Peshawar, Pakistan*
- CGPA: 3.30/4.0 (75.91%).
- Thesis: "Toward Reliable Fire & Smoke Instance Segmentation in Challenging Environments: A Vision-Based Approach."

SELECTED PUBLICATIONS

1. BoXFire: Geometry-Aware Box-Supervised Fusion for Accurate Fire and Smoke Segmentation. *Advanced Engineering Informatics*, 2026. **First author.**
2. Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey. *Computer Methods and Programs in Biomedicine*, 285, 109541, 2026. Co-author.
3. Privacy-Preserving Action Recognition: Taxonomy, Methods, and Privacy-Utility Trade-offs. arXiv:2608.04501, 2026. Co-author.
4. MedVLM: Vision-Language Model for Holistic Medical Applications. *IEEE Consumer Electronics Magazine*, 2025. **First author.**
5. **Box-Supervised Fire and Smoke Segmentation in Real-World Scenarios**. 21st International Conference on Multimedia Information Technology and Applications (MITA), 2025. **Best Paper Award**; press coverage.
6. WVES: A Weighted Ensemble Framework for Student Mental Health Assessment. *2026 International Conference on Technical Education (ICTechED)*.

RESEARCH COLLABORATIONS

Dr. Salman Khan, Research Fellow, Visual Artificial Intelligence Laboratory, Oxford Brookes University, UK

– Collaborated on cone detection, road-crack annotation, fire detection, and introductory 3D object detection using the KITTI dataset.

Dr. Mustaqeem Khan, Assistant Professor, United Arab Emirates University, UAE

– Collaborated on medical vision-language modeling and multimodal violence-detection experiments.

Dr. Alice Othmani, Postdoctoral Researcher, Ecole Normale Superieure de Paris, France

– Collaborated on Medical Question Answering: A Comprehensive Multimodal and LLM-Driven Survey.
– Research interests include pattern recognition, image and mesh analysis, contour processing, and shape-feature extraction (homepage).

ACHIEVEMENTS & SCHOLARSHIPS

– **Best Paper Award**, 21st MITA International Conference on Multimedia Information Technology and Applications, 2025 (press coverage).
– **Young Scientist Scholarship**, Chung-Ang University, South Korea, 2025 - Present.
– Fully funded Higher Education Commission scholarship, Pakistan, 2019 - 2023.
– Certificate in Artificial Intelligence, specializing in machine learning and deep learning, awarded by Pakistan's National Vocational and Technical Training Commission.

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, C++, C#, SQL
Python Libraries	OpenCV, Scikit-learn, Scikit-image, NumPy, Matplotlib
Deep Learning Frameworks	PyTorch, TensorFlow, Keras
Research Areas	Computer vision, multimodal AI, vision-language models, segmentation, medical AI

REFERENCES

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